

NMB	Rod End Bearing Division	Procedure	
Establish Date : 05 / SEP / 24	Standard for Engineering Change Control	No. PD-DV-EN-200000A	
Revision: [A] 13 / AUG / 25		Prepared by TH Engineer/QC/QA Approved by	Checked by Karuzawa Approved by
Effective Date : 22 / AUG / 25			
Occupation (ENG)			

1. PURPOSE

The purpose of this Engineering procedure is to establish a configuration management to ensure the identification and control of physical and functional attributes throughout the product lifecycle and guidelines for managing changes to Product & Manufacturing Design, Tooling Design, Machine or Inspection Program and applicable internal documents within Rod End Division. The manual aims to ensure that all changes are thoroughly evaluated, documented, approved, and implemented in a controlled and efficient manner, minimizing risks and disruptions to operations while maintaining product quality, reliability, and compliance with regulatory requirements.

2. SCOPE

The ECN process outlined in this manual applies to Manufacturing drawing changes, Tooling drawing changes and Program changes that impact the design, manufacturing, testing, or support of products, systems or managed by Rod End Division. The scope includes changes initiated internally and changes requested by customers or suppliers

3. DEPARTMENT IN CHARGE

Rod End Bearing Division. Engineering Department. shall be in charge.

4. REFERENCE SPECIFICATION

PD-DV-EN-300000	Standard for Document Control of Engineering
PD-DV-EN-400000	Standard for Process Control
PD-DV-QC-000069	Control of change
PD-DV-QC-000095	4M Change Request Standard
PD-DV-QC-940000	Control Manual for Identification, Traceability, and Configuration Management
PD-DV-QC-000060	Measurement System Analysis (MSA) Procedure

5. ENGINEERING CONTROL OF CHANGE

5.1 Type and reason of the change - Change Request Submission

The requester inputs the details through the Change Control System.

5.1.1 Changes by internal factors

This step involves the identification of the change request and the submission of the change proposal by ECR (Engineering Change Request) as show in Appendix 1.

5.1.2 Changes by external factors

This change is generally caused by customer's requirement change.

The review process is started from PD-DV-EN-300000 and then flown down to this process.

5.2 Preliminary Review by Engineering Department

The Engineering department conducts a preliminary review to ensure the change request is sufficient for the review

5.2.1 If the information is insufficient, the request can be rejected and sent back to the requester to add details and resubmit through the Change Control System.

5.2.2 If the request is deemed unfeasible or unreasonable, it can be denied or canceled with reasons provided to the requester.

UNCONTROLLED IF PRINTED

5.3 Impact Analysis, Risk Assessment and Notification to Customer

Engineering department shall identify the type of changes, impact to the product properties or interchangeability or significance of the process or method changes including inspections reference [Appendix 3](#), then ensure that the changes are managed through customer's QMS including public specification.

The proposed change shall be approved by customer before implementation of the change, if the customer approval of the change to the production are contractually required. reference PD-DV-QC-000069 Control of change.

If the evaluation shows an impact on the management and control of the four main factors (4M) that affect the production process results and the results of the product should be subject of verification and validation per PD-DV-EN-400000 appropriately when making an Engineering Change Notice refer PD-DV-QC-000095 4M Change Request Standard.

ECR arrangement shall be considering as follows;

5.3.1 ECN Issuance for Manufacturing Drawing Updates or Revisions.

In cases where updates or corrections to the manufacturing drawing are required, the Engineering Department shall proceed with issuing an Engineering Change Notice (ECN).

As part of this process, the status of the related drawing in the Innovator system shall be changed to "Suspend".

5.3.2 Reference Only Case - Test Result Recording for Evidence.

In certain cases, the change request may be treated as "Reference Only" for the purpose of recording test results as supporting evidence.

5.3.3 Product and Process authorized by Fujisawa plant or Karuizawa plant

The information shall be forwarded to KZW/FSW Engineer for further decision-making along with the testing results (if necessary).

5.4 Identifying Applicable Production Lots for the Change

The engineer checks related lots that need dispositioning by verifying and confirming the lot number, T/S, WIP, and stock. They manage cases where related lots need to be placed on 'HOLD' with PC or correct the drawing reference (PD-DV-QC-940000).

5.5 Plan and actions

Engineering shall create detailed plans, update, and register related documents/information, such as process drawings, tooling drawings, and machine programs.

Initiate notification to the QC function to evaluate and determine whether to conduct a Measurement System Analysis (MSA). If the decision is made not to perform an MSA, clear justification for the decision shall be provided. Refer PD-DV-QC-000060

If a First Article Inspection (FAI) is required, the traveler sheet should be stamped with "FAI Required" or have an "FA INSPECT" process added to the process router. Then performed and record the results.

5.6 Final Approval

The Engineering department manager conducts the final approval, including communicating ECN information to concerned sections if additional approval is needed.

5.7 Close change request

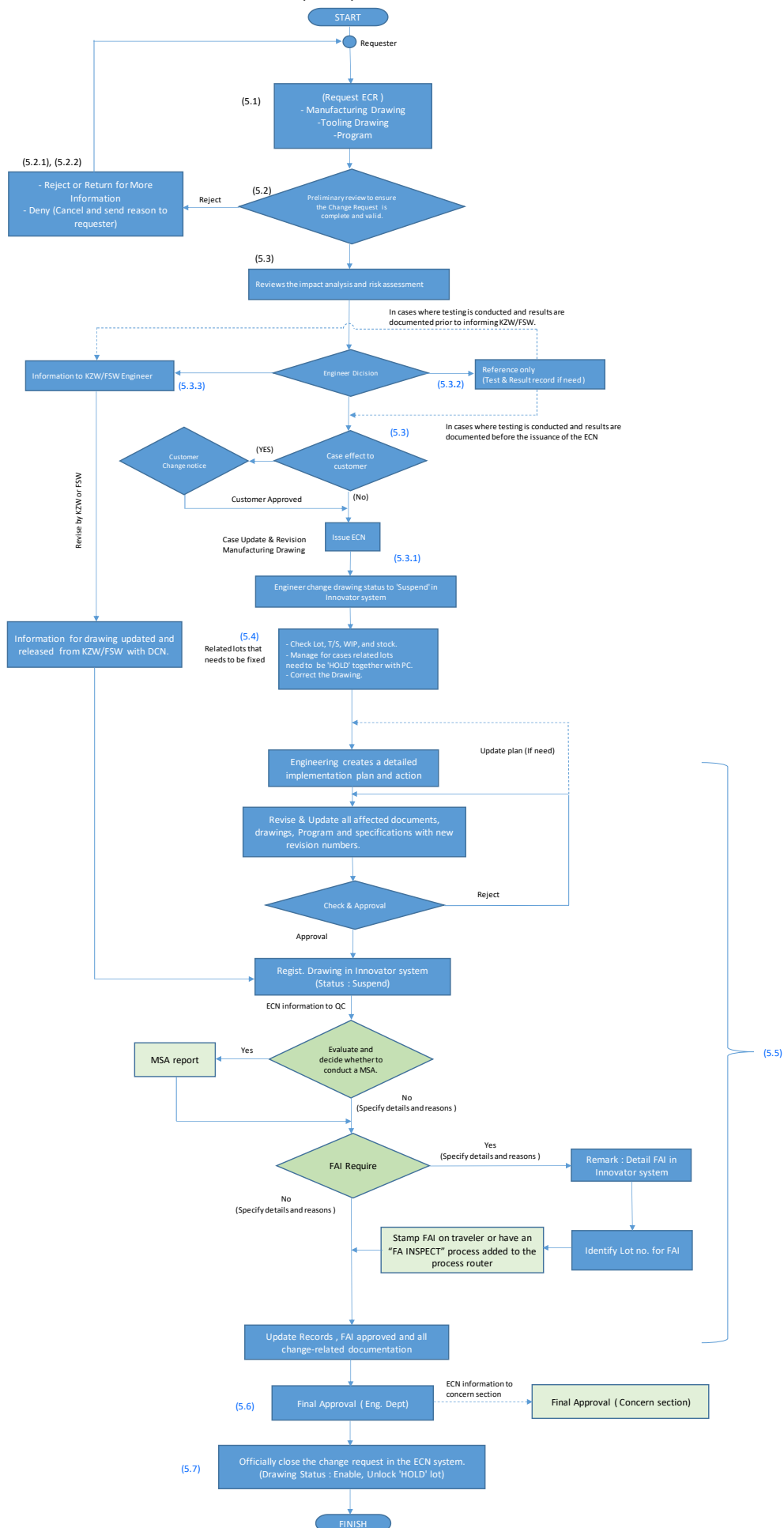
Officially close the change request in the Engineering change notice system and set the drawing status to 'Enable' and unlock the 'HOLD' lot. The details of the changes are summarized in the Engineering Change Notice (ECN) Form, as shown in the in [Appendix 2](#).

6. RECORD

The documents information shall be maintained per PD-DV-EN-300000

UNCONTROLLED IF PRINTED

ENGINEERING CHANGE NOTICE (ECN) PROCESS FLOW CHART.



UNCONTROLLED IF PRINTED

Appendix 1

Engineering Change Request (ECR) Form

ENGINEERING CHANGE REQUEST	
(FOR REQUESTER)	
REQUEST BY :	DATE OF REQUEST : DD/MM/YYYY
DEPARTMENT :	
TITLE OF CHANGE :	
REASON OF CHANGE :	
DETAILED DESCRIPTION OF CHANGE :	
BEFORE CHANGE :	AFTER CHANGE :
Notice! : This data is Reference Only	
☐ APPROVED REQUEST BY REQUESTER'S DEPARTMENT MANAGER	
SIGNATURE _____ DATE : DD/MM/YYYY	

* This document may be returned for further information if the information is incomplete.

(FOR Engineer Assigned)

Impact Assessment (การประเมินผลกระทบ) :

- ☐ **Customer**
- Customer approved refer document no. :
- ☐ **Sale Drawing**
- Update revision of Sale Drawing no. :
- ☐ **Traceability (track production history)**
- The delivered lot must be recalled and completed as necessary. : Lot no.
- ☐ **WIP, Stock, Back log**
- All lots in the Target Model List must be processed as specified in the ECN (See Target model list for product sheet).
- ☐ **Outsourcing process :**
- Notify the service provider. : Notification date

☐ By Document no.
☐ By Email

 - Ongoing lots (Lot no.) :
(Implement changes and Verify and inspect quality for lots can be updated or modified as required by the ECN.)
- ☐ **Unit Price**
- Inform the Purchase or stakeholders about the unit price change.
(In cases where changes increase production costs)
 - Factors contributing to increased production costs. :
- ☐ **Manufacturing process : Must be informed of the updated drawings to ensure proper manufacturing processes.**
 - Production methods and WI refer updated document no. :
 - Tooling data sheet refer updated document no. :
 - Program machine refer updated program no. :
 - Additional items to order related to ECN :
(e.g. tooling, M/C part)
- ☐ **Product quality & affect the ability to inspect or control quality : Must review the revised drawings to ensure that they meet internal and customer requirements.**
 - Inspection methods and WI refer updated document no. :
 - Additional items to order related to ECN :
(e.g. Quality Control Equipment, Calibration Tools)
- ☐ **Safety (increase the risk of accidents or damage)**
(e.g. Change program that increasing the operating speed of the machine without considering the impact on the safety of the parts or the machine)
 - Update/add risk assessment in the process refer updated doc no. :
- ☐ **On Time Delivery**
(e.g. New step in the manufacturing process that takes more time could extend the production timeline, New material with a longer lead time)
 - Increase approximate time :

(FOR Engineer Department Manager)

(FOR ENGINEER DEPARTMENT MANAGER)

- ☐ APPROVED ECR No. :

☐ DENY REASON :

☐ ISSUE ECN ☐ FOR REFERENCE ONLY
 REFER ECN No. :

Engineer Department manager

SIGNATURE

DATE :

DD/MM/YYYY

Remark : ECR No. Specified by

ECRT XX XX XX

Issue number in that month

Issue month, such as January, is set to 01

Issue year, such as 2524, is set to 24

UNCONTROLLED IF PRINTED

Appendix 2

Engineering Change Notice (ECN) Form

ENGINEERING CHANGE NOTICE			
(FOR ENGINEER ASSIGNED)			
ECN No. : FROM ECR No. : REFER DCN No. :			
REQUEST BY :	DATE OF REQUEST : DD/MM/YYYY		
DEPARTMENT :			
TITLE OF CHANGE :			
REASON OF CHANGE :			
Scope of Implementation (ขอบเขตการดำเนินการ) :			
BEFORE CHANGE :	AFTER CHANGE :		
<i>Notice! : This data is Reference Only</i>			
TARGET MODEL LIST ☐ The target model has been specified in the Target model list sheet ☐ Confirm Status Drawing in Innovator system : Suspend			
(FOR ENGINEER DEPARTMENT MANAGER)			
☐ APPROVED ECN (Start revise and update drawing, Document) ☐ DENY REASON : ☐ Return for more information :			
SIGNATURE			
DATE : DD/MM/YYYY			
(FOR PC, Engineer)			
Please check all relevant lots in WIP, Stock, Back Log and outsourcing processes to perform Hold lots. (Please refer Model in Target model list sheet)			
☐ The relevant lot has been specified in the Disposition lots list sheet ☐ No related lots			
(FOR QC : MSA)			
MSA evaluation by QC.			
☐ MSA require (Refer document No.) ☐ MSA not require			
Reason (QC) :			
SIGNATURE			
DATE : DD/MM/YYYY			
(FOR QC : FAI)			
QC Decision for FAI			
☐ Require Δ FAI ☐ Require Full FAI ☐ Not Require FAI			
Reason (QC) :			
SIGNATURE			
DATE : DD/MM/YYYY			
(FOR Engineer Assigned)			
Plan and action (Please see DWG. update status)			
FAI Lot No. :			
Related drawings have been revised. Drawing status in innovator system : Temporary			
Inform the concern department			
Need	No Need	PC Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	QA Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	QC Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	PD1 Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	PD2 Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	CM Dept INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
Need	No Need	MM Dept. INPUT NAME HERE	DATE : DD/MM/YYYY
☐	☐		
(FOR Engineer Assigned)			
☐ Confirm Drawing status in innovator system : ENABLE			
(FOR Engineer Department Manager)			
☐ Officially close the ECN (ECN EFFECTIVE)			
SIGNATURE			
DATE : DD/MM/YYYY			

Remark : ECN No. Generate by Innovator system. (Example ECNT101789)

UNCONTROLLED IF PRINTED

ROD END DIVISION, LOPBURI, NMB MINEBEA THAI LTD

Notice! : This data is Reference Only

Revision control sheet

Revision	Date	Detail	Approve
A	13 AUG 25	Topic 4 add PD-DV-QC-000060 Measurement System Analysis (MSA) Procedure Topic 5.1.1 Change "by ECR (Engineering Change Request system)" to "by ECR (Engineering Change Request) as show in Appendix 1." Remove topic 5.2.3 Topic 5.3 Change Appendix 2 to Appendix 3 Add topic 5.3.1, 5.3.2, 5.3.3 as follow: ECR arrangement shall be considering as follows; 5.3.1 ECN Issuance for Manufacturing Drawing Updates or Revisions. In cases where updates or corrections to the manufacturing drawing are required, the Engineering Department shall proceed with issuing an Engineering Change Notice (ECN). As part of this process, the status of the related drawing in the Innovator system shall be changed to "Suspend". 5.3.2 Reference Only Case - Test Result Recording for Evidence. In certain cases, the change request may be treated as "Reference Only" for the purpose of recording test results as supporting evidence. 5.3.3 Product and Process authorized by Fujisawa plant or Karuizawa plant The information shall be forwarded to KZW/FSW Engineer for further decision-making along with the testing results (if necessary). Remove the content in topic 5.5, replaced with the content of topic 5.6 and update to comply with the latest AS13100 requirements regarding MSA evaluation by QC Change 5.7 to 5.6 Change 5.8 to 5.7 and Appendix 1 to Appendix 2. Update ENGINEERING CHANGE NOTICE (ECN) PROCESS FLOW CHART refer update change. Change Appendix 1 from Example for Engineering Change Notice (ECN) Summary report to Engineering Change Request (ECR) Form Change Appendix 2 to Appendix 3 and replaced Appendix 2 by Engineering Change Notice (ECN) Form.	T.Sakda

UNCONTROLLED IF PRINTED